

Practical 10

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67.

The SSTF scheduling algorithm is used. The head is initially at cylinder

number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate

the total head movement (in number of cylinders) incurred while servicing these

requests.

```
tanuja@Tanuja: ~  
GNU nano 6.2 sstf.c *  
#include<stdio.h>  
#include<stdlib.h>  
  
int main()  
{  
    int request[8] = {98,183,41,122,14,124,65,67};  
    int visited[8] = {0};  
    int head = 53;  
    int total_movement = 0;  
  
    int i,j,min,distance,index;  
  
    for(i=0;i<8;i++)  
    {  
        min = 1000;  
  
        for(j=0;j<8;j++)  
        {  
            if(visited[j]==0)  
            {  
                distance = abs(head - request[j]);  
  
                if(distance < min)  
                {  
                    min = distance;  
                    index = j;  
                }  
            }  
        }  
  
        total_movement += min;  
        head = request[index];  
        visited[index] = 1;  
    }  
  
    printf("Total Head Movement = %d\n", total_movement);  
}
```

Output:

```
tanuja@Tanuja:~$ nano sstf.c
tanuja@Tanuja:~$ gcc sstf.c
tanuja@Tanuja:~$ ./a.out
Total Head Movement = 323
tanuja@Tanuja:~$ |
```